PARTNERSHIP MEMO

# United States–Korea Strategic Energy & Industrial Partnership

Where deeper industrial integration creates strategic value, through which instruments, and where to stay out

*Generated from Allied Industrial Base Explorer data and a curated external evidence layer (accessed 2026-11-06); evidence, sources and caveats in the appendix. Data as of 30 October 2026.*

## Executive Decision Summary

**BUILD TOGETHER**

Battery Materials & Cells: Natural graphite (joint); Battery recycling feed (US->Korea); Accumulator parts & separators (Korea->US). Nuclear Fuel & Reactor Systems: Nuclear extraction (joint). Permanent Magnets & Drive Systems: Rare-earth oxides & compounds (US->Korea); Rare-earth permanent magnets (joint); AC generators (US->Korea). Grid Equipment & Conductors: Copper smelting & refining (joint). Solar Manufacturing: Polysilicon (US->Korea).

**TRADE AND STANDARDS**

Recognise trusted partner supply of electric vehicles refining (rare-earth metals, rare-earth oxides & compounds); rare-earth oxides & compounds; accumulator parts & separators in procurement and content rules. Common anti-circumvention rules for rare-earth permanent magnets. Bind market access for wind components (ac generators, ferrite & other magnets, insulators & bushings). Keep uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores); nuclear special materials; transformers open to global sourcing.

**INSTITUTIONS**

Upgrade US-Korea Strategic Investment MOU and Special Act (US$350bn): sector strategy, reciprocal commitments, a KPI dashboard and an annual review. No new MOU.

**STAY OUT**

Do not co-subsidise new Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) capacity. Do not co-subsidise new Batteries assembly (li-ion cells & packs, other accumulators) capacity. Do not co-subsidise new Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks) capacity.

**Why this partnership matters.**

- Of 142 technology x stage cells and electro-industrial nodes assessed for the pair, 39 show an active complementarity (a supplier meeting the other's need, a shared chokepoint, a capacity-constrained trusted supplier or a chain specialisation), 3 direct competition and 74 no strategic case. The relationship is not one of general similarity: most of the electro-industrial base needs no bilateral intervention.[1]
- AIB strategic roles: US Demand & Market Anchor (primary), Coordination / Bridge Hub (primary), Capital & Investment Anchor (primary), Scale Manufacturing & Integration Platform (secondary), Specialized Technology & Manufacturing Anchor (secondary); Korea Specialized Technology & Manufacturing Anchor (primary), Scale Manufacturing & Integration Platform (primary), Coordination / Bridge Hub (secondary), Capital & Investment Anchor (secondary).[1]

**The central bilateral bargain.**

- *Inference.* Korea supplies accumulator parts & separators, solar cells, cathode materials & precursors, gallium, germanium & indium where the US is short; the US supplies polysilicon, rare-earth oxides & compounds, battery recycling feed, ac generators where Korea is short; the two share exposure to the same coercive supplier in natural graphite, rare-earth permanent magnets, copper smelting & refining, where neither can supply the other and the answer is joint demand, finance and third-country capacity.
- *Inference.* From the AIB roles, the US brings demand and offtake, coordination across allied chains, capital; Korea brings specialised technology and manufacturing, scale manufacturing, coordination across allied chains. Public finance runs through DOE LPO, DoW OSC, DFC, EXIM on the US side and KEXIM, K-SURE, KOMIR, KDB on the Korea side. The trade is market access and demand certainty for trusted capacity.

**The 5 highest-priority joint bets** (tier 1 by the rules in Section 3, one per compact; ranked by urgency then importance):

| PRIORITY | SECTOR | NODES IN THE BET (DIRECTION) | STRATEGIC PROBLEM | US CONTRIBUTION | KOREA CONTRIBUTION | MAIN INSTRUMENT (LEAD NODE) | TIMING |
|---|---|---|---|---|---|---|---|
| 1 | Battery Materials & Cells | Natural graphite (joint); Battery recycling feed (US->Korea); Accumulator parts & separators (Korea->US) | geopolitical chokepoint | Natural graphite: its demand for pooled offtake, and finance for non-Chinese supply; Battery recycling feed: trusted capacity (44% of world supply) | Natural graphite: its demand for pooled offtake, and finance for non-Chinese supply; Battery recycling feed: demand and long-term offtake | anchor offtake | 12-24 months |
| 2 | Nuclear Fuel & Reactor Systems | Nuclear extraction (joint) | geopolitical chokepoint | Nuclear extraction: its demand for pooled offtake, and finance for non-Chinese supply | Nuclear extraction: its demand for pooled offtake, and finance for non-Chinese supply | anchor offtake | 12-24 months |
| 3 | Permanent Magnets & Drive Systems | Rare-earth oxides & compounds (US->Korea); Rare-earth permanent magnets (joint); AC generators (US->Korea) | geopolitical chokepoint | Rare-earth oxides & compounds: trusted capacity (5% of world supply); Rare-earth permanent magnets: its demand for pooled offtake, and finance for non-Chinese supply | Rare-earth oxides & compounds: demand and long-term offtake; Rare-earth permanent magnets: its demand for pooled offtake, and finance for non-Chinese supply | anchor offtake | 12-24 months |
| 4 | Grid Equipment & Conductors | Copper smelting & refining (joint) | geopolitical chokepoint | Copper smelting & refining: its demand for pooled offtake, and finance for non-Chinese supply; Solar refining: its demand for pooled offtake, and finance for non-Chinese supply | Copper smelting & refining: its demand for pooled offtake, and finance for non-Chinese supply; Solar refining: its demand for pooled offtake, and finance for non-Chinese supply | anchor offtake | 12-24 months |
| 5 | Solar Manufacturing | Polysilicon (US->Korea) | geopolitical chokepoint | Polysilicon: trusted capacity (31% of world supply) | Polysilicon: demand and long-term offtake | strategic imports | 12-24 months |

Sources: 1.

- *Inference.* Outside the electro-industrial compacts, the rules also find fuel-trade lanes (oil extraction); they are handled in the trade agenda, not as compacts.

**Do not prioritise** (Section 10):

- *Recommendation.* Do not co-subsidise new Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) capacity: both are strong suppliers (US 7%, Korea 2%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* Do not co-subsidise new Batteries assembly (li-ion cells & packs, other accumulators) capacity: both are strong suppliers (US 5%, Korea 5%). More trusted capacity is needed, so agree who specialises in what rather than racing.
- *Recommendation.* Do not co-subsidise new Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks) capacity: both are strong suppliers (US 8%, Korea 4%). Trusted supply is adequate: this is commercial competition.

**Existing mechanisms and the MEA question** (Section 8):

- *Inference.* MEA assessment: **upgrade an existing framework (USKR-M2)**. Gaps: no cross-sector KPI dashboard or review cycle; no bilateral mechanism covers SC-BATTERIES, SC-MAGNETS, SC-GRID.
- *Recommendation.* Do not sign a new energy MOU. Give USKR-M2 a sector strategy, reciprocal commitments and a KPI dashboard covering the recommended compacts, reviewed every 12 months.

## 1. Current State of the Relationship

This section sets out what already exists before anything new is proposed. Facts are from the curated evidence layer (accessed 2026-11-06); items marked partial or unverified rest on secondary sources and should be confirmed before citation.

**Alliance and plurilateral architecture**

- Korea's memberships in OSI's economic-security agreements table: IPEF, MSP, U.S. FTA.[1]
- Senior officials meeting in New York agreed to enhance dialogue on critical minerals and cooperation in Southeast Asia and Sub-Saharan Africa through the trilateral Early Warning System.[2]
- Fact sheet states US support for a process toward Korean civil enrichment and reprocessing, subject to the 123 Agreement and US law, and approves nuclear-powered submarines; formal talks reportedly began June 2026 and were still discussing 123 revision in September 2026; no revision signed.[3]
- ASPI lists the MSP, which Korea chairs in 2025, among the frameworks for Australia-Korea minerals cooperation; Australian membership is not stated in that article.[4]
- The US-hosted ministerial in Feb 2026 was attended by Japan and Korea among others, and Korea was reported to chair the FORGE critical-minerals forum through June 2026.[5]
- Korea chaired the Minerals Security Partnership until June 2026; a February 2026 report says its 17 members carried into the successor FORGE; Korean public lenders joined the MSP Finance Network; no signed US-Korea critical-minerals MOU was found.[6]

**Trade architecture and frictions**

- In the OSI energy basket, US imported US$41.2bn from Korea and Korea imported US$33.1bn from US (bilateral flows 2024, summed over the technology x stage cells).[1]
- KORUS has been in force since March 2012; amended in 2018 on autos, light trucks and investment rules without congressional action; the countries still use its consultative mechanisms; exact day of entry into force not stated in the source.[7]
- Westinghouse, KEPCO and KHNP settled their APR1400 intellectual-property dispute on 16 Jan 2025, dismissing all legal actions and setting up future cooperation on projects worldwide; terms are confidential, so market-allocation reports cannot be confirmed.[8]

- ICE detained 475 people at the US$4.3bn Georgia plant in September 2025, many Korean technicians on visa issues; Hyundai said opening slipped two to three months and reports in April 2026 said it was ready to open.[9]
- A January 2026 Section 301 petition targets Korea's treatment of Coupang, on which USTR has not acted; a July 2026 House Judiciary interim report called it discriminatory; the US also criticises the pace of Korean investment; reported January 2026 threat of 25% tariffs not in this source.[10]

**Industrial and energy agreements**

- The three leaders agreed to pilot an early warning system exchanging supply-chain information via embassies in selected countries, with critical minerals and rechargeable batteries as priority products, plus an annual commerce and industry ministers track; the system is not published as operating beyond the pilot.[11]
- Sets a 15% US tariff on Korean goods (higher of KORUS, MFN or 15%), 15% Section 232 on autos and parts, no more than 15% on pharma, semiconductors no less favourable than comparable partners; commits US$350bn (US$200bn plus US$150bn shipbuilding) with a US$20bn annual cap; reproduced text, not the whitehouse.gov page.[12]
- Lutnick and Minister Kim signed the MOU on the US$200bn strategic-investment tranche, capped at US$20bn a year; reported profit split is 50:50 until principal and interest are repaid, then 10:90 in the US favour.[13]

**Investment vehicle, project pipeline and financing**

- The US$150bn shipbuilding half of the pledge is to be financed through direct investment, guarantees and loans led by Korean policy-finance institutions; reports say Hanwha Philly Shipyard expansion is delayed because the fund has not taken concrete shape; article date not shown.[14]
- Joint statement of 1 Oct 2026 formalises three projects: Encinal, Texas 6.3 GW gas plant (US$22.3bn), Alaska LNG (over US$50bn expected, subject to commercial reasonableness) and nuclear (up to US$120bn); the annual cap and vehicle structure are not addressed in this article.[15]
- Up to US$120bn for eight US reactors (US$100bn construction, US$20bn contingency), with a US$10bn advance payment for long-lead items targeted by year-end 2026; links to the May 2025 goal of 10 large reactors under construction by 2030; reactor designs not specified in this article.[16]
- Korean companies will pursue a significant minority stake in Westinghouse to share gains from large-reactor projects, with terms to be set in commercial talks; other reports put it at 5 to 10 percent but that figure is not in this source.[17]

**Corporate relationships in the recommended compacts**

- LGES agreed to supply Toyota's North American EVs with about 20 GWh a year of modules from 2025, with roughly 4 trillion won (about 3 billion dollars) invested in Michigan lines dedicated to Toyota.[18]
- SK On signed a conditional offtake with US graphite developer Westwater for battery-grade natural graphite in North America.[19]
- Multi-metal smelter project of about US$7.4bn through Crucible JV, with the Department of War reported as largest shareholder at 40% (US$760m in one account, US$1.4bn conditional in another) and a US$210m Commerce grant.[20]
- The 2023 Honda-LGES joint venture L-H Battery keeps operating its Ohio cell plant under a lease after LGES sold the building to Honda for about 4.22 trillion won (2.85 billion dollars) announced Dec 2025; a later report of a lower closing value of 3.74 trillion won is unconfirmed.[21]
- LGES agreed to sell the Ohio JV factory building and related assets to Honda for about $2.85B with the JV continuing via lease and closing targeted for 2026-02-28.[22]
- A US$200m JV, run by POSCO International as largest shareholder with ReElement refining technology, plans US NdPr-Dy-Tb oxides then magnets and motor cores, following a September 2025 MOU; no MP Materials link found.[23]

- Qcells began cell manufacturing at Cartersville, Georgia; by Q3 2026 it expects 3.3 GW each of ingot, wafer and cell and 3.5 GW of modules there, plus 5.1 GW modules at Dalton; ingot and wafer status not stated; investment not disclosed; a secondary report cites US$675m of 2026 AMPC.[24]
- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[25]
- LGES began mass production of LFP storage cells by repurposing part of its Holland, Michigan EV plant, targeting over 30 GWh by end-2026; a US$1.6bn LFP supply deal with DTE Energy is reported; publication date approximate.[26]
- Samsung SDI secured a KRW 1.5tn (about US$1bn) US ESS cell supply deal, starting with NCA before LFP; its Kokomo, Indiana line was reported starting ESS output in 2026; the StarPlus (Stellantis) and BlueOval SK joint ventures were not covered; date approximate.[27]
- Second Montgomery, Alabama plant (312,000 sq ft, about US$200m) due for completion in April and to lift extra-high-voltage transformer capacity about 50%, up to 765 kV; completion year inferred; other outlets give US$273.7m across Ulsan and Alabama.[28]
- Hyosung is investing about US$300m in a three-phase expansion of its ultra-high-voltage transformer plant in Memphis, Tennessee, and formed a JV with Quanta Services in June 2026 for extra-high-voltage circuit breakers; completion dates not given.[29]
- LS Electric plans US$240m of US investment by 2030, expanding its Utah switchgear plant and developing a Bastrop, Texas campus.[30]
- Unverified items are left out of this section; they stay in the evidence register (evidence_register.csv) and the evidence layer.[1]
- *Inference.* Working channels today: US-Korea Strategic Investment MOU and Special Act (US$350bn) (a project pipeline and finance); Minerals Security Partnership and FORGE (finance). What the relationship lacks: no cross-sector KPI dashboard or review cycle; no bilateral mechanism covers SC-BATTERIES, SC-MAGNETS, SC-GRID.

## 2. What Each Country Is Good At — and What It Needs

Cell codes: position / need for **US · Korea**. Position S strong (top quarter of contenders and above the capability floor), M contender, W not a contender; need H high (supply-chain need >= 0.67, or deployment need >= 0.60 at Deployment), L low; - not measured. Source: strategic_position_tbl (engine choices are in country_profiles.csv).

| TECHNOLOGY | EXTRACTION | REFINING | COMPONENTS | ASSEMBLY | DEPLOYMENT | RECOVERY |
|---|---|---|---|---|---|---|
| Batteries | W/H · W/H | M/L · W/H | S/L · M/H | S/H · S/H | S/H · S/L | -/H · -/H |
| Coal | S/L · W/H | M/H · W/H | M/L · S/L | M/L · S/L | S/L · S/L | -/- · -/- |
| Electric Grid | M/H · W/H | W/H · W/H | M/L · M/H | M/L · M/L | -/L · -/L | -/- · -/- |
| Electric Vehicles | M/H · W/H | M/H · W/H | S/H · S/H | S/L · S/L | S/L · S/H | -/- · -/- |
| Gas | S/L · W/H | S/L · W/H | S/L · M/L | M/L · S/L | S/H · S/L | -/- · -/- |
| Geothermal | -/- · -/- | -/- · -/- | -/- · -/- | -/- · -/- | S/L · M/H | -/- · -/- |
| Green Hydrogen | M/- · W/- | W/- · W/- | -/- · -/- | -/- · -/- | -/L · -/H | -/- · -/- |
| Nuclear | W/H · W/H | W/H · W/H | M/L · S/L | M/L · S/L | S/L · S/H | -/- · -/- |
| Oil | S/L · W/H | S/L · M/L | M/L · M/H | M/L · S/L | S/L · S/L | -/H · -/L |
| Solar | M/H · W/H | W/H · W/H | S/L · M/L | S/L · M/L | S/L · M/H | -/- · -/- |

| TECHNOLOGY | EXTRACTION | REFINING | COMPONENTS | ASSEMBLY | DEPLOYMENT | RECOVERY |
|---|---|---|---|---|---|---|
| Wind | S/- · W/- | M/H · W/H | S/L · M/H | S/L · M/L | S/L · S/H | -/- · -/- |

Sources: 1.

**Electro-industrial nodes where either country has a position or a need** (AIB division of labour, 2024 trade; share = world export or production share; need is the node's technology x stage need, inherited, so a proxy; demand = share of allied demand). 74 of 76 nodes:

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

*Table A1 in the appendix (74 rows) carries this table in full; the CSV outputs carry every column.*

- Revealed policy priority (NIPO / IEA PAMS within-country percentile; policy never enters the classification) is highest for US in: Green Hydrogen assembly (0.95); Green Hydrogen components (0.93); Nuclear components (0.92); Nuclear assembly (0.90); Batteries assembly (0.87); for Korea in: Batteries assembly (0.96); Green Hydrogen assembly (0.95); Nuclear assembly (0.92); Nuclear components (0.90); Electric Vehicles recovery (0.87).[1]
- Evidence quality: 41 of 66 cells have high stage-specific evidence for both countries; 19 units are flagged low-evidence.[1]

## 3. Bilateral Complementarity Map

State counts over all units: a_supplies_b_needs 8; complementary_specialization 14; shared_vulnerability 8; capacity_constrained_trusted_supply 9; direct_competition 3; low_strategic_case 74; deployment_demand 8; not_assessed 18.[1]

*Inference.* Rules (Section 3 of docs/partnership_memos.md): two strong positions are **direct competition**, never complementarity; a supplier must be strong (or a contender the engine names as the other's trusted partner) and the buyer must have high need, or material demand where trusted capacity is short or a chokepoint exists; **capacity-constrained** where the AIB finds allied capacity short of demand; **shared vulnerability** where both need the node, neither supplies it and a coercive supplier holds at least half of world supply.

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

*Table A2 in the appendix (42 rows) carries this table in full; the CSV outputs carry every column.*

- Low-priority / open-market nodes (56): wafers, synthetic graphite (anode-grade), pv modules & generators, li-ion cells & packs, rare-earth metals, uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), uranium conversion & enrichment, nuclear reactors & enrichment machinery, fuel assemblies & zirconium, reactor parts, electric & plug-in cars, electric motors, ferrite & other magnets, motor & generator parts, electric two-wheelers & works trucks, rare-earth ores (monazite), cobalt intermediates & chemicals, wind turbine generating sets, lithium chemicals, pv cover glass, wind blades & hubs, wind towers, lithium minerals, battery materials (foil, electrolyte), other accumulators, manganese dioxide (emd), nickel intermediates & chemicals, cobalt ores, piezo-electric quartz, heat pumps & ac units, static converters & inverters, power electronics components, solar thermal water heaters, grain-oriented electrical steel (goes), electrolysers, refrigeration & heat-pump compressors, power semiconductors, insulators & bushings, electricity meters, electric furnaces & heating, furnace parts, transformers, switchgear & protection, hydro turbine parts, graphite electrodes, blade core wood (balsa), copper ores & concentrates, copper wire & rod, nuclear special materials, overhead conductors, power cables, aluminium wire rod, direct-reduced iron (dri/hbi), hydrogen, ammonia & methanol, pgms (iridium, ruthenium), hydro turbines.[1]
- Deployment cells are demand context, not supply relations: high deployment need in Green Hydrogen (KOR); Nuclear (KOR); Electric Vehicles (KOR); Solar (KOR); Wind (KOR); Geothermal (KOR); Batteries (USA); Gas (USA).[1]

## 4. Trade Agenda

- *Inference.* **What the US should export to Korea:** Polysilicon; Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds); Rare-earth oxides & compounds; Battery recycling feed; Wind components (ac generators, ferrite & other magnets, insulators & bushings); Batteries components (accumulator parts & separators); AC generators; Oil extraction; Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite); Electric buses, trucks & tractors; Gas components; Gas extraction; Gas refining; Refrigerants; Sensors & controller parts; Automatic controllers; Semiconductor manufacturing equipment; Silver powder; Oil refining; Coal extraction.
- *Inference.* **What Korea should export to the US:** Accumulator parts & separators; Solar cells; Cathode materials & precursors; Gas assembly; Gallium, germanium & indium; Non-oriented electrical steel (NGO); Integrated circuits; Oil assembly (ac generators); Electric rail vehicles; Coal components; Coal assembly.
- *Inference.* **Where tariffs would undermine allied specialisation:** Wind|Components.
- *Inference.* **Procurement, standards and content rules to recognise mutually:** Electric Vehicles|Refining, rare-earth-oxides-and-compounds, accumulator-parts-and-separators.
- *Inference.* **Where third-country circumvention safeguards are genuinely required:** rare-earth-permanent-magnets; elsewhere they would add cost without security.
- *Inference.* **Where unrestricted global sourcing is acceptable:** uranium-fuel-upstream--aggregated, nuclear-special-materials, transformers, uranium-conversion-and-enrichment, overhead-conductors, wind-towers.

*Table A3 in the appendix (21 rows) carries this table in full; the CSV outputs carry every column.*

- The tariff measures in force or in dispute between the two, the baseline any binding of the nodes above works from:
- Korean autos and parts face a 15% Section 232 rate (combined with MFN) under the November 2025 deal, applied retroactively to 1 Nov 2025 for autos per implementing guidance; Section 232 is unaffected by the IEEPA ruling.[31]
- The deal caps future Section 232 chip tariffs for Korea, subject to talks; trade press reports a January 2026 25% Section 232 tariff on certain advanced chips with Korean memory exports to data centres largely exempt.[32]
- After the Supreme Court ruling the President lifted the EO 14257 tariffs, including Korea's 15% country rate, on 20 Feb 2026 and imposed a temporary 10% global tariff under Section 122, which expired in July 2026; refund treatment not covered.[33]
- On 23 Jul 2026 the US imposed a 12.5% tariff on most Korean goods after a Section 301 forced-labor investigation; a separate excess-capacity Section 301 probe could add tariffs and Seoul wants assurance the total stays within the agreed 15%.[34]
- Proclamation of 6 Aug 2026 sets a 15% tariff on certain derivatives and minimum import prices on polysilicon and derivatives from 4 Dec 2026; Korea's combined rate is capped at 15% under the deal; dates and scope taken from secondary summaries.[35]
- Tariffs on most Korean steel and aluminium imports remain at 50%; the fact sheet does not address them and the earlier steel quota was terminated in 2025.[36]

## 5. Investment Agenda

### US → Korea investment (US capital and firms expanding in Korea)

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

| ID | NODE | CAPITAL / CAPABILITY OWNER | DESTINATION | DEMAND CASE | PRIVATE ACTORS (EVIDENCE) | PUBLIC INSTRUMENT | ADDITIONALITY | HORIZON |
|---|---|---|---|---|---|---|---|---|
| I07 | Polysilicon | USA | KOR | KOR takes 10% of allied demand (8% of allied net imports); AIB class LOW INTERVENTION | | strategic imports | medium | 2-3 years |
| I08 | Wind components (ac generators, ferrite & other magnets, insulators & bushings) | USA | KOR | supply-chain need 0.27 (USA) / 0.80 (KOR) | | market access | medium | 2-3 years |
| I09 | Batteries components (accumulator parts & separators) | USA | KOR | supply-chain need 0.06 (USA) / 0.73 (KOR) | | export credit | medium | 2-3 years |
| I10 | AC generators | USA | KOR | KOR takes 8% of allied demand (8% of allied net imports); AIB class LOW INTERVENTION | | export credit | medium | 2-3 years |
| I11 | Electric buses, trucks & tractors | USA | KOR | KOR takes 3% of allied demand (3% of allied net imports); AIB class BUILD | | anchor offtake | medium | 2-3 years |
| I12 | Refrigerants | USA | KOR | KOR takes 5% of allied demand (5% of allied net imports); AIB class BUILD | | anchor offtake | high | 2-3 years |
| I13 | Sensors & controller parts | USA | KOR | KOR takes 4% of allied demand (1% of allied net imports); AIB class BUILD | | standards alignment | high | 2-3 years |
| I14 | Automatic controllers | USA | KOR | KOR takes 6% of allied demand (2% of allied net imports); AIB class LOW INTERVENTION | | standards alignment | medium | 2-3 years |
| I15 | Semiconductor manufacturing equipment | USA | KOR | KOR takes 27% of allied demand (16% of allied net imports); AIB class LOW INTERVENTION | | standards alignment | medium | 2-3 years |
| I16 | Silver powder | USA | KOR | KOR takes 16% of allied demand (12% of allied net imports); AIB class LOW INTERVENTION | | standards alignment | medium | 2-3 years |

Sources: 1.

## Korea → US investment (Korea capital and firms expanding in US)

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

| ID | NODE | CAPITAL / CAPABILITY OWNER | DESTINATION | DEMAND CASE | PRIVATE ACTORS (EVIDENCE) | PUBLIC INSTRUMENT | ADDITIONALITY | HORIZON |
|---|---|---|---|---|---|---|---|---|
| I01 | Battery recycling feed | KOR | USA | KOR takes 20% of allied demand (20% of allied net imports); AIB class LOW INTERVENTION | Toyota Tsusho; LG Energy Solution | strategic imports | medium | 2-3 years |

| ID | NODE | CAPITAL / CAPABILITY OWNER | DESTINATION | DEMAND CASE | PRIVATE ACTORS (EVIDENCE) | PUBLIC INSTRUMENT | ADDITIONALITY | HORIZON |
|---|---|---|---|---|---|---|---|---|
| I02 | Accumulator parts & separators | KOR | USA | USA takes 42% of allied demand (39% of allied net imports); AIB class BUILD | | anchor offtake | medium | 2-3 years |
| I03 | Oil extraction | KOR | USA | supply-chain need 0.47 (USA) / 0.99 (KOR) | | anchor offtake | medium | 2-3 years |
| I04 | Cathode materials & precursors | KOR | USA | USA takes 5% of allied demand (0% of allied net imports); AIB class BUILD | | standards alignment | high | 2-3 years |
| I05 | Gallium, germanium & indium | KOR | USA | USA takes 42% of allied demand (32% of allied net imports); AIB class BUILD | | anchor offtake | high | 2-3 years |
| I06 | Non-oriented electrical steel (NGO) | KOR | USA | USA takes 0% of allied demand (0% of allied net imports); AIB class BUILD | | standards alignment | high | 2-3 years |

Sources: 1.

### Joint investment (at home or in third countries)

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

| ID | NODE | CAPITAL / CAPABILITY OWNER | DESTINATION | DEMAND CASE | PRIVATE ACTORS (EVIDENCE) | PUBLIC INSTRUMENT | ADDITIONALITY | HORIZON |
|---|---|---|---|---|---|---|---|---|
| I17 | Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds) | USA+KOR | third countries (THA; MYS; JPN) and either partner | supply-chain need 0.90 (USA) / 0.99 (KOR) | POSCO International; ReElement | anchor offtake | medium | 2-3 years |
| I18 | Rare-earth oxides & compounds | USA+KOR | third countries (JPN; MYS) and either partner | KOR takes 7% of allied demand (4% of allied net imports); AIB class BUILD | POSCO International; ReElement | anchor offtake | medium | 2-3 years |
| I19 | Solar cells | USA+KOR | third countries (MYS; THA) and either partner | USA takes 40% of allied demand (39% of allied net imports); AIB class BUILD | Hanwha Qcells | anchor offtake | low | 2-3 years |
| I20 | Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite) | USA+KOR | third countries (CHL; ARG; BRA) and either partner | supply-chain need 0.82 (USA) / 1.00 (KOR) | Arafura Rare Earths; SK On; Westwater Resources | anchor offtake | low | 2-3 years |

Sources: 1.

- *Inference.* Additionality = mean of three named components: the market failure (chokepoint 1, capacity constraint 0.75, gap or emerging bottleneck 0.5, none 0), one minus an existing commitment in the evidence layer (committed 1, announced 0.5) and one minus a working market (the buyer already takes 30% or more of its imports from the supplier). It is **low** whenever there is no failure or the investment is already committed: public support would then not be additional.

## 6. Third-Country and Development Cooperation

- From the development-partnership pairs (scripts/93: a donor's need meets a developing economy's credible Catch Up pathway in the same step), US and Korea are both donors at 10 recipient steps. Ranked: both donors' own need first, then steps inside a recommended compact.[1]
- *Inference.* Roles: US brings demand and offtake and coordination across compacts; Korea brings specialised technology and components and equipment and manufacturing at scale (AIB strategic roles).

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

| ID | RECIPIENT | TECHNOLOGY | STAGE | COMPACT | RECIPIENT CHOICE | RECIPIENT ANCHOR | US INTEREST | KOREA INTEREST |
|---|---|---|---|---|---|---|---|---|
| X01 | Malaysia | Electric Vehicles | Refining | SC-MAGNETS | catch up | direct supply share; direct export share | own secure need (USA) | own secure need (KOR) |
| X02 | Thailand | Electric Vehicles | Refining | SC-MAGNETS | catch up | direct export share | own secure need (USA) | own secure need (KOR) |
| X03 | Indonesia | Electric Vehicles | Components | SC-EV; SC-MAGNETS | preserve | direct supply share; direct export share; adjacent stage share | own preserve need (USA) | own preserve need (KOR) |
| X04 | Thailand | Electric Vehicles | Components | SC-EV; SC-MAGNETS | catch up | direct supply share; direct export share | own preserve need (USA) | own preserve need (KOR) |
| X05 | Philippines | Electric Vehicles | Components | SC-EV; SC-MAGNETS | catch up | direct supply share; direct export share | own preserve need (USA) | own preserve need (KOR) |
| X06 | Viet Nam | Batteries | Assembly | SC-BATTERIES | catch up | direct supply share; direct export share | own preserve need (USA) | own preserve need (KOR) |
| X07 | Malaysia | Electric Vehicles | Components | SC-EV; SC-MAGNETS | catch up | direct export share | own preserve need (USA) | own preserve need (KOR) |
| X08 | Indonesia | Batteries | Assembly | SC-BATTERIES | catch up | direct supply share | own preserve need (USA) | own preserve need (KOR) |
| X09 | Malaysia | Batteries | Assembly | SC-BATTERIES | catch up | direct supply share; direct export share | own preserve need (USA) | own preserve need (KOR) |
| X10 | Mexico | Electric Vehicles | Components | SC-EV; SC-MAGNETS | catch up | direct supply share; direct export share; adjacent stage share | own preserve need (USA) | own preserve need (KOR) |

Sources: 1.

- The three leaders agreed to pilot an early warning system exchanging supply-chain information via embassies in selected countries, with critical minerals and rechargeable batteries as priority products, plus an annual commerce and industry ministers track; the system is not published as operating beyond the pilot.[11]
- LGES agreed to supply Toyota's North American EVs with about 20 GWh a year of modules from 2025, with roughly 4 trillion won (about 3 billion dollars) invested in Michigan lines dedicated to Toyota.[18]
- Senior officials meeting in New York agreed to enhance dialogue on critical minerals and cooperation in Southeast Asia and Sub-Saharan Africa through the trilateral Early Warning System.[2]
- ASPI lists the MSP, which Korea chairs in 2025, among the frameworks for Australia-Korea minerals cooperation; Australian membership is not stated in that article.[4]
- The 2023 Honda-LGES joint venture L-H Battery keeps operating its Ohio cell plant under a lease after LGES sold the building to Honda for about 4.22 trillion won (2.85 billion dollars) announced Dec 2025; a later report of a lower closing value of 3.74 trillion won is unconfirmed.[21]
- LGES agreed to sell the Ohio JV factory building and related assets to Honda for about $2.85B with the JV continuing via lease and closing targeted for 2026-02-28.[22]

- The US-hosted ministerial in Feb 2026 was attended by Japan and Korea among others, and Korea was reported to chair the FORGE critical-minerals forum through June 2026.[5]
- Korea chaired the Minerals Security Partnership until June 2026; a February 2026 report says its 17 members carried into the successor FORGE; Korean public lenders joined the MSP Finance Network; no signed US-Korea critical-minerals MOU was found.[6]
- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[25]

## 7. Recommended Sectoral Compacts

*Inference.* 5 compacts pass (rule: at least one joint bet, at least two active nodes and both countries holding an AIB compact role; or, labelled **evidence-led**, an existing bilateral pipeline or at least 2 bilateral project items where OSI measures a need but cannot see the partner's capability). The hypothesis test for every AIB compact:

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

| RANK | COMPACT | VERDICT | JOINT BETS | COORDINATE | US ROLE | KOREA ROLE | BILATERAL MECHANISMS | RECOMMENDED |
|---|---|---|---|---|---|---|---|---|
| 1 | Battery Materials & Cells | bilateral priority | 3 | 1 | Resource supplier | Specialized technology supplier | JPKR-M6; USKR-M5 | TRUE |
| 2 | Permanent Magnets & Drive Systems | bilateral priority | 3 | 1 | Scale manufacturer | Specialized technology supplier | JPKR-M6 | TRUE |
| 3 | Solar Manufacturing | bilateral priority | 1 | 2 | Demand anchor | Specialized technology supplier | USKR-M3; USKR-M5 | TRUE |
| 4 | Grid Equipment & Conductors | evidence-led candidate | 1 | 0 | Demand anchor | Scale manufacturer | USKR-M5 | TRUE |
| 5 | Nuclear Fuel & Reactor Systems | evidence-led candidate | 0 | 0 | Specialized technology supplier | Scale manufacturer | USKR-M2; USKR-M4 | TRUE |
| 6 | Electric Vehicles | secondary | 0 | 2 | Scale manufacturer | Scale manufacturer | USKR-M3 | FALSE |
| 7 | Heat Pumps & Cooling | secondary | 0 | 1 | Processing anchor | Scale manufacturer | | FALSE |
| 8 | Semiconductors & Power Electronics | secondary | 0 | 5 | Specialized technology supplier | Specialized technology supplier | USKR-M3 | FALSE |
| 9 | Industrial Electrification & Hydrogen | not a bilateral priority | 0 | 0 | Demand anchor | Scale manufacturer | USKR-M3 | FALSE |
| 10 | Wind & Hydropower Equipment | not a bilateral priority | 0 | 0 | Demand anchor | Specialized technology supplier | | FALSE |

Sources: 1.

### 1. Battery Materials & Cells

#### Strategic objective

- AIB mission: Build missing capacity in Li-ion cells & packs and lithium chemicals and manganese dioxide (EMD) (refined chemicals); targeted: preserve and second-source specialized production in lithium chemicals (refined chemicals)..[1]
- *Recommendation.* Change the outcome at natural graphite, battery recycling feed, accumulator parts & separators: geopolitical chokepoint; emerging bottleneck; trusted partner capacity constraint.

**Why this pair?**

- *Inference.* Natural graphite: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 82% of world supply). Battery recycling feed: US supplies / Korea needs (USA is a capable trusted supplier and the other side needs the node). Accumulator parts & separators: Capacity-constrained trusted supply (KOR is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Cathode materials & precursors: Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: US Resource supplier (lead node battery-recycling-feed); Korea Specialized technology supplier (lead node cathode-materials-and-precursors).[1]

**Supply-chain architecture**

| STAGE | UNIT | BASIS | US POSITION | KOREA POSITION | DESIRED BILATERAL ARRANGEMENT | TREATMENT | INSTRUMENT |
|---|---|---|---|---|---|---|---|
| Extraction | natural-graphite | AIB node | moderate | moderate | shared vulnerability | BUILD | anchor offtake |
| Refining | cathode-materials-and-precursors | AIB node | moderate | strong | complementary specialization | COORDINATE | standards alignment |
| Components | accumulator-parts-and-separators | AIB node | moderate | strong | capacity constrained trusted supply | BUILD | anchor offtake |
| Assembly | li-ion-cells-and-packs | AIB node | moderate | moderate | low strategic case | LOW INTERVENTION | |
| Deployment | Batteries/Deployment | OSI cell (no AIB node at this stage) | strong | strong | deployment demand | DEMAND CONTEXT | |
| Recovery | battery-recycling-feed | AIB node | strong | weak | a supplies b needs | COORDINATE | strategic imports |

Sources: 1.

**Reciprocal bargain**

- US provides: supply of battery recycling feed; long-term demand and offtake for accumulator parts & separators, cathode materials & precursors; procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Korea provides: capacity and technology in accumulator parts & separators, cathode materials & precursors; long-term demand for battery recycling feed; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: pooled offtake and finance for non-coercive natural graphite.
- Third-country partners needed: MDG, MOZ, DEU, HUN, POL, JPN.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MDG at natural graphite; DEU at synthetic graphite (anode-grade); POL at li-ion cells & packs; FIN at cobalt intermediates & chemicals; CHL at lithium chemicals; TWN at battery materials (foil, electrolyte); VNM at other accumulators; JPN at manganese dioxide (emd). The bilateral compact should not try to replace them there.[1]

**Policy instruments (minimum necessary)**

- *Recommendation.* anchor_offtake for natural graphite; strategic_imports for battery recycling feed; anchor_offtake for accumulator parts & separators; standards_alignment for cathode materials & precursors

**Priority transactions**

- *Recommendation.* Long-term offtake by US and Korea buyers for non-coercive natural graphite, sized to underwrite new capacity (candidate hosts: MDG, MOZ, DEU) (D02).
- *Recommendation.* Multi-year supply agreements for US battery recycling feed into Korea buyers (D06).

**Success by 2030**

- Natural graphite: coercive share of supply to the two buyers below 50% (CHN now holds 82% of world supply).
- Battery recycling feed: allied residual gap under the China shock closed (now 8% of allied demand); US's share of Korea's imports up from 4%.
- Accumulator parts & separators: allied residual gap under the China shock closed (now 37% of allied demand); Korea's share of US's imports up from 21%.
- Cathode materials & precursors: coercive share of supply to the two buyers below 50% (CHN now holds 92% of world supply); Korea's share of US's imports up from 6%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

**Failure / exit conditions**

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

**Evidence (external)**

- The three leaders agreed to pilot an early warning system exchanging supply-chain information via embassies in selected countries, with critical minerals and rechargeable batteries as priority products, plus an annual commerce and industry ministers track; the system is not published as operating beyond the pilot.[11]
- LGES agreed to supply Toyota's North American EVs with about 20 GWh a year of modules from 2025, with roughly 4 trillion won (about 3 billion dollars) invested in Michigan lines dedicated to Toyota.[18]
- SK On signed a conditional offtake with US graphite developer Westwater for battery-grade natural graphite in North America.[19]
- A Korean report says the US has restricted exports of black mass and tungsten scrap since 27 Aug 2025, so US-origin recycling feed is unavailable to Korean recyclers.[37]
- ICE detained 475 people at the US$4.3bn Georgia plant in September 2025, many Korean technicians on visa issues; Hyundai said opening slipped two to three months and reports in April 2026 said it was ready to open.[9]
- Senior officials meeting in New York agreed to enhance dialogue on critical minerals and cooperation in Southeast Asia and Sub-Saharan Africa through the trilateral Early Warning System.[2]
- POSCO Future M began cathode exports to the US in October 2025 and its GM joint venture Ultium CAM is in Quebec; no Korean cathode plant on US soil was confirmed in this search; L&F targets North American LFP supply by end Q3 2026; EcoPro not found.[38]
- The 2023 Honda-LGES joint venture L-H Battery keeps operating its Ohio cell plant under a lease after LGES sold the building to Honda for about 4.22 trillion won (2.85 billion dollars) announced Dec 2025; a later report of a lower closing value of 3.74 trillion won is unconfirmed.[21]

- Section 45X advanced-manufacturing credit remains available to Korean plants but foreign-entity sourcing limits have applied since 1 Jan 2026, pushing Korean makers toward non-Chinese materials.[39]
- LGES agreed to sell the Ohio JV factory building and related assets to Honda for about $2.85B with the JV continuing via lease and closing targeted for 2026-02-28.[22]
- LGES began mass production of LFP storage cells by repurposing part of its Holland, Michigan EV plant, targeting over 30 GWh by end-2026; a US$1.6bn LFP supply deal with DTE Energy is reported; publication date approximate.[26]
- Samsung SDI secured a KRW 1.5tn (about US$1bn) US ESS cell supply deal, starting with NCA before LFP; its Kokomo, Indiana line was reported starting ESS output in 2026; the StarPlus (Stellantis) and BlueOval SK joint ventures were not covered; date approximate.[27]

## 2. Permanent Magnets & Drive Systems

### Strategic objective

- AIB mission: Build missing capacity in rare-earth permanent magnets (magnets & core steel)..[1]
- *Recommendation.* Change the outcome at rare-earth oxides & compounds, rare-earth permanent magnets, ac generators: geopolitical chokepoint; strategic manufacturing gap.

### Why this pair?

- *Inference.* Rare-earth oxides & compounds: Capacity-constrained trusted supply (USA is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Rare-earth permanent magnets: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 64% of world supply). AC generators: US supplies / Korea needs (USA is a capable trusted supplier and the other side needs the node). Non-oriented electrical steel (NGO): Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: US Scale manufacturer (lead node ac-generators); Korea Specialized technology supplier (lead node non-oriented-electrical-steel-ngo).[1]

### Supply-chain architecture

| STAGE | UNIT | BASIS | US POSITION | KOREA POSITION | DESIRED BILATERAL ARRANGEMENT | TREATMENT | INSTRUMENT |
|---|---|---|---|---|---|---|---|
| Extraction | rare-earth-ores-monazite | AIB node | moderate | weak | low strategic case | LOW INTERVENTION | |
| Refining | rare-earth-oxides-and-compounds | AIB node | moderate | weak | capacity constrained trusted supply | BUILD | anchor offtake |
| Components | rare-earth-permanent-magnets | AIB node | moderate | weak | shared vulnerability | BUILD | anchor offtake |
| Assembly | ac-generators | AIB node | strong | weak | a supplies b needs | COORDINATE | strategic imports |
| Deployment | Electric Vehicles/Deployment | OSI cell (no AIB node at this stage) | strong | strong | deployment demand | DEMAND CONTEXT | |
| Recovery | Electric Vehicles/Recovery | OSI cell (no AIB node at this stage) | not measured | not measured | not assessed | INSUFFICIENT DATA | |

Sources: 1.

**Reciprocal bargain**

- US provides: supply of rare-earth oxides & compounds, ac generators; long-term demand and offtake for non-oriented electrical steel (ngo); procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Korea provides: capacity and technology in non-oriented electrical steel (ngo); long-term demand for rare-earth oxides & compounds, ac generators; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: pooled offtake and finance for non-coercive rare-earth permanent magnets.
- Third-country partners needed: JPN, MYS, PHL, VNM, TWN.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MYS at rare-earth oxides & compounds; JPN at rare-earth permanent magnets; VNM at rare-earth metals; DEU at electric motors; DEU at ferrite & other magnets; DEU at motor & generator parts. The bilateral compact should not try to replace them there.[1]

**Policy instruments (minimum necessary)**

- *Recommendation.* anchor_offtake for rare-earth oxides & compounds; anchor_offtake for rare-earth permanent magnets; strategic_imports for ac generators; standards_alignment for non-oriented electrical steel (ngo)

**Priority transactions**

- *Recommendation.* Long-term offtake by US and Korea buyers for non-coercive rare-earth permanent magnets, sized to underwrite new capacity (candidate hosts: JPN, PHL, VNM) (D04).
- *Recommendation.* Tariff binding for US wind components (ac generators, ferrite & other magnets, insulators & bushings) entering Korea (D05).

**Success by 2030**

- Rare-earth oxides & compounds: coercive share of supply to the two buyers below 50% (CHN now holds 84% of world supply); US's share of Korea's imports up from 12%.
- Rare-earth permanent magnets: coercive share of supply to the two buyers below 50% (CHN now holds 64% of world supply).
- AC generators: allied residual gap under the China shock closed (now 4% of allied demand); US's share of Korea's imports up from 20%.
- Non-oriented electrical steel (NGO): allied residual gap under the China shock closed (now 17% of allied demand); Korea's share of US's imports up from 1%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

**Failure / exit conditions**

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

**Evidence (external)**

- The US-hosted ministerial in Feb 2026 was attended by Japan and Korea among others, and Korea was reported to chair the FORGE critical-minerals forum through June 2026.[5]
- A US$200m JV, run by POSCO International as largest shareholder with ReElement refining technology, plans US NdPr-Dy-Tb oxides then magnets and motor cores, following a September 2025 MOU; no MP Materials link found.[23]

- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[25]

### 3. Solar Manufacturing

#### Strategic objective

- AIB mission: Build missing capacity in modules, cells & cover glass and polysilicon..[1]
- *Recommendation.* Change the outcome at polysilicon, solar cells, silver powder: geopolitical chokepoint; strategic manufacturing gap.

#### Why this pair?

- *Inference.* Polysilicon: US supplies / Korea needs (USA is a capable trusted supplier and the other side needs the node). Solar cells: Capacity-constrained trusted supply (KOR is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Silver powder: Complementary specialisation (USA is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: US Demand anchor (lead node solar-cells); Korea Specialized technology supplier (lead node solar-cells).[1]

#### Supply-chain architecture

| STAGE | UNIT | BASIS | US POSITION | KOREA POSITION | DESIRED BILATERAL ARRANGEMENT | TREATMENT | INSTRUMENT |
|---|---|---|---|---|---|---|---|
| Extraction | Solar/Extraction | OSI cell (no AIB node at this stage) | moderate | weak | low strategic case | LOW INTERVENTION | |
| Refining | polysilicon | AIB node | strong | moderate | a supplies b needs | COORDINATE | strategic imports |
| Components | solar-cells | AIB node | moderate | strong | capacity constrained trusted supply | BUILD | anchor offtake |
| Assembly | pv-modules-and-generators | AIB node | moderate | weak | low strategic case | LOW INTERVENTION | |
| Deployment | Solar/Deployment | OSI cell (no AIB node at this stage) | strong | moderate | deployment demand | DEMAND CONTEXT | |
| Recovery | Solar/Recovery | OSI cell (no AIB node at this stage) | not measured | not measured | not assessed | INSUFFICIENT DATA | |

Sources: 1.

#### Reciprocal bargain

- US provides: supply of polysilicon, silver powder; long-term demand and offtake for solar cells; procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Korea provides: capacity and technology in solar cells; long-term demand for polysilicon, silver powder; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: common specifications and qualification.
- Third-country partners needed: MYS, JPN, DEU, THA.

- Better placed than this pair (largest trusted supplier once China is removed, AIB): JPN at wafers; VNM at pv modules & generators; MYS at pv cover glass. The bilateral compact should not try to replace them there.[1]

**Policy instruments (minimum necessary)**

- *Recommendation.* strategic_imports for polysilicon; anchor_offtake for solar cells; standards_alignment for silver powder

**Priority transactions**

- *Recommendation.* Multi-year supply agreements for US polysilicon into Korea buyers (D07).
- *Recommendation.* Long-term offtake by US buyers for Korea solar cells, sized to underwrite new capacity (candidate hosts: MYS, THA) (D09).

**Success by 2030**

- Polysilicon: coercive share of supply to the two buyers below 50% (CHN now holds 97% of world supply); US's share of Korea's imports up from 20%.
- Solar cells: coercive share of supply to the two buyers below 50% (CHN now holds 92% of world supply); Korea's share of US's imports up from 8%.
- Silver powder: allied residual gap under the China shock closed (now 0% of allied demand); US's share of Korea's imports up from n/a.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

**Failure / exit conditions**

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

**Evidence (external)**

- Qcells began cell manufacturing at Cartersville, Georgia; by Q3 2026 it expects 3.3 GW each of ingot, wafer and cell and 3.5 GW of modules there, plus 5.1 GW modules at Dalton; ingot and wafer status not stated; investment not disclosed; a secondary report cites US$675m of 2026 AMPC.[24]
- Proclamation of 6 Aug 2026 sets a 15% tariff on certain derivatives and minimum import prices on polysilicon and derivatives from 4 Dec 2026; Korea's combined rate is capped at 15% under the deal; dates and scope taken from secondary summaries.[35]
- Korea's trade ministry asked Washington for flexible treatment in the polysilicon probe, saying Hanwha Qcells and OCI avoid cheap Chinese polysilicon, with Qcells sourcing mainly from OCI's Malaysian plants; US-to-Korea polysilicon volumes are not documented in this source.[40]

### 4. Grid Equipment & Conductors (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Korea world share: copper smelting & refining 1.7%, grain-oriented electrical steel (goes) 5.2%, insulators & bushings 0.8%, electricity meters 0.0%, transformers 9.8%, switchgear & protection 3.7%, copper ores & concentrates 0.0%, copper wire & rod 1.3%, overhead conductors 1.2%, power cables 8.4%, aluminium wire rod 0.0%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.
- Need that OSI does measure in the compact: copper smelting & refining (US high 0.90, 15% of allied demand; Korea high 0.96); insulators & bushings (US demand 0.32, 19% of allied demand; Korea low (cell 0.80; not a material buyer)); electricity meters (US demand 0.33, 22% of allied demand; Korea low 0.64); transformers (US demand 0.33, 40% of allied demand; Korea low 0.64); switchgear & protection (US demand 0.33, 20% of allied

demand; Korea low 0.64); copper ores & concentrates (US low (cell 0.89; not a material buyer), 0% of allied demand; Korea high 0.97); copper wire & rod (US high 0.96, 10% of allied demand; Korea low (cell 0.96; not a material buyer)); overhead conductors (US demand 0.33, 32% of allied demand; Korea low 0.64); power cables (US demand 0.33, 20% of allied demand; Korea low 0.64); aluminium wire rod (US demand n/a, 24% of allied demand; Korea not_measured n/a).[1]

### Strategic objective

- AIB mission: Monitor: no intervention class reaches the materiality floor (25% of the compact's supply to shift or US$1bn); targeted: build missing capacity in grain-oriented electrical steel (GOES) (wire rod & transformer steel)..[1]
- *Recommendation.* Change the outcome at copper smelting & refining: geopolitical chokepoint.

### Why this pair?

- *Inference.* Copper smelting & refining: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 50% of world supply).
- AIB compact roles: US Demand anchor (lead node transformers); Korea Scale manufacturer (lead node transformers).[1]

### Supply-chain architecture

| STAGE | UNIT | BASIS | US POSITION | KOREA POSITION | DESIRED BILATERAL ARRANGEMENT | TREATMENT | INSTRUMENT |
|---|---|---|---|---|---|---|---|
| Extraction | copper-ores-and-concentrates | AIB node | moderate | weak | low strategic case | LOW INTERVENTION | |
| Refining | copper-smelting-and-refining | AIB node | weak | weak | shared vulnerability | BUILD | anchor offtake |
| Components | insulators-and-bushings | AIB node | moderate | weak | low strategic case | LOW INTERVENTION | |
| Assembly | transformers | AIB node | moderate | moderate | low strategic case | LOW INTERVENTION | |
| Deployment | Electric Grid/Deployment | OSI cell (no AIB node at this stage) | not measured | not measured | low strategic case | LOW INTERVENTION | |
| Recovery | Electric Grid/Recovery | OSI cell (no AIB node at this stage) | not measured | not measured | not assessed | INSUFFICIENT DATA | |

Sources: 1.

### Reciprocal bargain

- US provides: procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Korea provides: procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: pooled offtake and finance for non-coercive copper smelting & refining.
- Third-country partners needed: none identified by the AIB host model.

- Better placed than this pair (largest trusted supplier once China is removed, AIB): JPN at grain-oriented electrical steel (goes); ITA at insulators & bushings; MEX at electricity meters. The bilateral compact should not try to replace them there.[1]

**Policy instruments (minimum necessary)**

- *Recommendation.* anchor_offtake for copper smelting & refining

**Priority transactions**

- *Recommendation.* Long-term offtake by US and Korea buyers for non-coercive solar refining (copper wire & rod, polysilicon), sized to underwrite new capacity (candidate hosts: MYS, JPN, DEU) (D03).

**Success by 2030**

- Copper smelting & refining: coercive share of supply to the two buyers below 50% (CHN now holds 50% of world supply).
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

**Failure / exit conditions**

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

**Evidence (external)**

- Multi-metal smelter project of about US$7.4bn through Crucible JV, with the Department of War reported as largest shareholder at 40% (US$760m in one account, US$1.4bn conditional in another) and a US$210m Commerce grant.[20]
- Second Montgomery, Alabama plant (312,000 sq ft, about US$200m) due for completion in April and to lift extra-high-voltage transformer capacity about 50%, up to 765 kV; completion year inferred; other outlets give US$273.7m across Ulsan and Alabama.[28]
- Hyosung is investing about US$300m in a three-phase expansion of its ultra-high-voltage transformer plant in Memphis, Tennessee, and formed a JV with Quanta Services in June 2026 for extra-high-voltage circuit breakers; completion dates not given.[29]
- LS Electric plans US$240m of US investment by 2030, expanding its Utah switchgear plant and developing a Bastrop, Texas campus.[30]

## 5. Nuclear Fuel & Reactor Systems (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Korea world share: uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores) 0.0%, uranium conversion & enrichment 0.0%, nuclear reactors & enrichment machinery 0.0%, fuel assemblies & zirconium 7.4%, reactor parts 0.0%, nuclear special materials 0.1%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.
- Need that OSI does measure in the compact: uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores) (US high 0.99, 91% of allied demand; Korea low (cell 0.96; not a material buyer)); uranium conversion & enrichment (US high 0.99, 42% of allied demand; Korea high 0.98); nuclear reactors & enrichment machinery (US demand 0.47, 8% of allied demand; Korea low 0.62); nuclear special materials (US demand n/a, 51% of allied demand; Korea demand n/a).[1]

**Strategic objective**

- AIB mission: Build missing capacity in reactors & enrichment machinery and nuclear special materials (fuel & nuclear materials); coordinate trade to redirect existing supply in reactor components; preserve and second-source specialized production in reactors & enrichment machinery..[1]
- *Recommendation.* Use USKR-M2; USKR-M4 to tie the announced projects to the nodes where OSI measures need, and verify the partner capability that trade data cannot see (ownership abroad, domestic sales, heavy components) before any public finance is added.

**Why this pair?**

- *Inference.* No OSI-observed complementarity; see the evidence below.
- AIB compact roles: US Specialized technology supplier (lead node reactor-parts); Korea Scale manufacturer (lead node fuel-assemblies-and-zirconium).[1]

**Supply-chain architecture**

| STAGE | UNIT | BASIS | US POSITION | KOREA POSITION | DESIRED BILATERAL ARRANGEMENT | TREATMENT | INSTRUMENT |
|---|---|---|---|---|---|---|---|
| Extraction | uranium-fuel-upstream--aggregated | AIB node | weak | weak | low strategic case | LOW INTERVENTION | |
| Refining | uranium-conversion-and-enrichment | AIB node | moderate | weak | low strategic case | LOW INTERVENTION | |
| Components | reactor-parts | AIB node | strong | weak | low strategic case | LOW INTERVENTION | |
| Assembly | nuclear-reactors-and-enrichment-machinery | AIB node | weak | weak | low strategic case | LOW INTERVENTION | |
| Deployment | Nuclear/Deployment | OSI cell (no AIB node at this stage) | strong | strong | deployment demand | DEMAND CONTEXT | |
| Recovery | Nuclear/Recovery | OSI cell (no AIB node at this stage) | not measured | not measured | not assessed | INSUFFICIENT DATA | |

Sources: 1.

**Reciprocal bargain**

- US provides: procurement recognition of the partner as trusted supply; finance for the compact's nodes (DOE LPO, DoW OSC, DFC, EXIM).
- Korea provides: procurement recognition of the partner as trusted supply; finance for capacity in the partner market (KEXIM, K-SURE, KOMIR, KDB).
- Jointly: common specifications and qualification.
- Third-country partners needed: none identified by the AIB host model.

**Policy instruments (minimum necessary)**

- *Recommendation.* Use the existing investment vehicle; no new instrument.

**Priority transactions**

- *Recommendation.* Long-term offtake by US and Korea buyers for non-coercive nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores)), sized to

underwrite new capacity (D01).

**Success by 2030**

- Projects in the pipeline reach final investment decision with partner-made equipment in the bill of materials.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

**Failure / exit conditions**

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

**Evidence (external)**

- Westinghouse, KEPCO and KHNP settled their APR1400 intellectual-property dispute on 16 Jan 2025, dismissing all legal actions and setting up future cooperation on projects worldwide; terms are confidential, so market-allocation reports cannot be confirmed.[8]
- Fact sheet states US support for a process toward Korean civil enrichment and reprocessing, subject to the 123 Agreement and US law, and approves nuclear-powered submarines; formal talks reportedly began June 2026 and were still discussing 123 revision in September 2026; no revision signed.[3]
- Up to US$120bn for eight US reactors (US$100bn construction, US$20bn contingency), with a US$10bn advance payment for long-lead items targeted by year-end 2026; links to the May 2025 goal of 10 large reactors under construction by 2030; reactor designs not specified in this article.[16]
- Korean companies will pursue a significant minority stake in Westinghouse to share gains from large-reactor projects, with terms to be set in commercial talks; other reports put it at 5 to 10 percent but that figure is not in this source.[17]

## 8. Master Energy Agreement Assessment

- *Inference.* Rule: **existing** if every recommended compact is covered by a mechanism with both a pipeline and finance and a cross-sector mechanism has KPIs and a review cycle; **upgrade** if at least 50% of compacts are covered by a mechanism with finance or a pipeline, or a cross-sector mechanism with a pipeline exists; **new MEA** otherwise.
- *Inference.* Result: **upgrade**; coverage 20% of recommended compacts; cross-sector mechanism with KPIs and review: no.

| COMPACT | ANY MECHANISM | FINANCE OR PIPELINE | PIPELINE AND FINANCE |
|---|---|---|---|
| SC-BATTERIES | FALSE | FALSE | FALSE |
| SC-MAGNETS | FALSE | FALSE | FALSE |
| SC-SOLAR | TRUE | FALSE | FALSE |
| SC-GRID | FALSE | FALSE | FALSE |
| SC-NUCLEAR | TRUE | TRUE | TRUE |

Sources: 1.

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

| ID | MECHANISM | STATUS | SCOPE | PIPELINE | FINANCE | KPIS | REVIEW | RECIPROCAL |
|---|---|---|---|---|---|---|---|---|
| JPKR-M6 | US-Japan-ROK trilateral supply-chain early warning and critical-minerals dialogue | announced | SC-BATTERIES/SC-MAGNETS | FALSE | FALSE | FALSE | partial | TRUE |
| USKR-M1 | KORUS Free Trade Agreement | in force | cross sector | FALSE | FALSE | FALSE | TRUE | TRUE |
| USKR-M2 | US-Korea Strategic Investment MOU and Special Act (US$350bn) | in force | cross sector/SC-NUCLEAR | TRUE | TRUE | FALSE | partial | partial |
| USKR-M3 | US tariff arrangement and its legal basis (15% deal, IEEPA ruling, Section 122/301/232) | in force | SC-EV/SC-SEMIS/SC-SOLAR/SC-INDUSTRY | FALSE | FALSE | FALSE | partial | FALSE |
| USKR-M4 | US-Korea civil nuclear cooperation (123 revision, Westinghouse-KHNP, reactor build-out) | proposed | SC-NUCLEAR | partial | partial | FALSE | partial | partial |
| USKR-M5 | Government-backed Korean plants in US supply chains (Korea Zinc Crucible smelter, Qcells, battery and grid plants) | announced | SC-SOLAR/SC-BATTERIES/SC-GRID | partial | partial | FALSE | FALSE | partial |
| USKR-M6 | Minerals Security Partnership and FORGE | in force | cross sector | FALSE | partial | FALSE | partial | TRUE |

Sources: 1.

**Recommended: upgrade US-Korea Strategic Investment MOU and Special Act (US$350bn) into the execution framework for the compacts**

- **Scope:** Battery Materials & Cells; Permanent Magnets & Drive Systems; Solar Manufacturing; Grid Equipment & Conductors; Nuclear Fuel & Reactor Systems; no other sectors.
- **Reciprocal commitments:** US commits demand certainty (procurement recognition and anchor offtake) and market-access predictability for the nodes Korea supplies (accumulator parts & separators, solar cells, cathode materials & precursors, non-oriented electrical steel (ngo)); Korea commits the same for the nodes US supplies (polysilicon, rare-earth oxides & compounds, battery recycling feed, ac generators). Each commits public finance for the compact nodes, at home, in the partner or with named third-country hosts.
- **Compact portfolio:** Battery Materials & Cells (bilateral priority); Permanent Magnets & Drive Systems (bilateral priority); Solar Manufacturing (bilateral priority); Grid Equipment & Conductors (evidence-led candidate); Nuclear Fuel & Reactor Systems (evidence-led candidate).
- **Financing lanes:** US: DOE LPO, DoW OSC, DFC, EXIM; Korea: KEXIM, K-SURE, KOMIR, KDB; demand-side tools (offtake, stockpiles, procurement): DOE, DoW OSC, DLA and MCEE, KOMIR, PPS, NSSC; third-country finance: DFC and KEXIM, KOMIR, KOICA.
- **Governance:** one bilateral steering line inside US-Korea Strategic Investment MOU and Special Act (US$350bn). Projects are selected by a joint committee, not by one side.
- **Milestones:** the 12-month milestones of the deal pipeline (Section 9).
- **KPI dashboard:** per compact node, the coercive share of each country's supply, the bilateral import share, offtake signed (volume), capacity at FID and operating, and public money committed vs private.
- **Review cycle:** every 12 months, with the OSI tables re-run on new trade data.
- **Sunset / exit:** each compact carries the exit conditions in Section 7; the framework expires with the investment vehicle () unless renewed on evidence.

## 9. 12–36 Month Deal Pipeline

*Inference.* Ranked lexicographically on separate fields (importance, then additionality, urgency, tractability, execution speed, value-chain interdependency, evidence quality); urgency is not importance. At most two deals per

compact.

*Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.*

| RANK | DEAL / ACTION | COMPACT | INSTRUMENT | RESPONSIBLE INSTITUTIONS | 12M MILESTONE | IMPORTANCE | URGENCY | ADDITIONALITY |
|---|---|---|---|---|---|---|---|---|
| 1 | Long-term offtake by US and Korea buyers for non-coercive nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores)), sized to underwrite new capacity | SC-NUCLEAR | anchor offtake | USA: DOE, DoW OSC, DLA; KOR: MCEE, PPS | Signed long-term offtake from at least one large buyer | high | high | high |
| 2 | Long-term offtake by US and Korea buyers for non-coercive natural graphite, sized to underwrite new capacity (candidate hosts: MDG, MOZ, DEU) | SC-BATTERIES | anchor offtake | USA: DOE, DoW OSC, DLA; KOR: MCEE, PPS | Signed long-term offtake from at least one large buyer | high | high | high |
| 3 | Long-term offtake by US and Korea buyers for non-coercive solar refining (copper wire & rod, polysilicon), sized to underwrite new capacity (candidate hosts: MYS, JPN, DEU) | SC-GRID | anchor offtake | USA: DOE, DoW OSC, DLA; KOR: MCEE, PPS | Signed long-term offtake from at least one large buyer | high | high | high |
| 4 | Long-term offtake by US and Korea buyers for non-coercive rare-earth permanent magnets, sized to underwrite new capacity (candidate hosts: JPN, PHL, VNM) | SC-MAGNETS | anchor offtake | USA: DOE, DoW OSC, DLA; KOR: MCEE, PPS | Signed long-term offtake from at least one large buyer | high | high | high |
| 5 | Tariff binding for US wind components (ac generators, ferrite & other magnets, insulators & bushings) entering Korea | SC-MAGNETS | market access | USA: USTR, Commerce; KOR: MOTIR (MOTIE), K-SURE, KOTRA | Tariff treatment of the node bound or exempted for the trusted supplier | high | high | medium |
| 6 | Multi-year supply agreements for US battery recycling feed into Korea buyers | SC-BATTERIES | strategic imports | USA: USTR, Commerce; KOR: MOTIR (MOTIE), K-SURE, KOTRA | Named import lanes and volumes agreed between buyers and the supplier's producers | high | high | medium |
| 7 | Multi-year supply agreements for US polysilicon into Korea buyers | SC-SOLAR | strategic imports | USA: USTR, Commerce; KOR: MOTIR (MOTIE), K-SURE, KOTRA | Named import lanes and volumes agreed between buyers and the supplier's producers | high | high | medium |
| 8 | Long-term offtake by Korea buyers for US oil extraction, sized to underwrite new capacity | | anchor offtake | USA: DOE, DoW OSC, DLA; KOR: MCEE, PPS | Signed long-term offtake from at least one large buyer | high | low | medium |
| 9 | Long-term offtake by US buyers for Korea solar cells, sized to underwrite new capacity (candidate hosts: MYS, THA) | SC-SOLAR | anchor offtake | USA: DOE, DoW OSC, DLA; KOR: MCEE, PPS | Signed long-term offtake from at least one large buyer | high | high | low |
| 10 | Long-term offtake by US buyers for Korea gallium, germanium & indium, sized to underwrite new capacity (candidate hosts: SGP) | SC-SEMIS | anchor offtake | USA: DOE, DoW OSC, DLA; KOR: MCEE, PPS | Signed long-term offtake from at least one large buyer | medium | low | high |

| RANK | DEAL / ACTION | COMPACT | INSTRUMENT | RESPONSIBLE INSTITUTIONS | 12M MILESTONE | IMPORTANCE | URGENCY | ADDITIONALITY |
|---|---|---|---|---|---|---|---|---|
| 11 | Long-term offtake by Korea buyers for US refrigerants, sized to underwrite new capacity (candidate hosts: NLD) | SC-HEAT | anchor offtake | USA: DOE, DoW OSC, DLA; KOR: MCEE, PPS | Signed long-term offtake from at least one large buyer | medium | low | high |
| 12 | Joint specification work plan for semiconductor manufacturing equipment | SC-SEMIS | standards alignment | USA: USTR, NRC; KOR: MOTIR (MOTIE), MCEE, NSSC | Joint list of specifications that fragment the market and a work plan to align them | medium | low | medium |

Sources: 1.

## 10. What Not to Do

- *Recommendation.* **N01** Do not co-subsidise new Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) capacity: both are strong suppliers (US 7%, Korea 2%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N02** Do not co-subsidise new Batteries assembly (li-ion cells & packs, other accumulators) capacity: both are strong suppliers (US 5%, Korea 5%). More trusted capacity is needed, so agree who specialises in what rather than racing.
- *Recommendation.* **N03** Do not co-subsidise new Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks) capacity: both are strong suppliers (US 8%, Korea 4%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N04** Do not create a new bilateral energy MOU or a free-standing MEA institution: USKR-M2 already exists (finance and a project pipeline); dock the compacts in it.
- *Recommendation.* **N05** Do not put new public money behind Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite) investment that is already committed (evidence JPKR-11; USAU-09; USKR-29); support is additional only beyond it.
- *Recommendation.* **N06** Do not put new public money behind Solar cells investment that is already committed (evidence USKR-21); support is additional only beyond it.

## 11. Evidence Register

- Every recommendation (IDs J, C, T, I, X, D, N, M) has rows in evidence_register.csv: the OSI metrics behind it (variable, value for each country, year, source, evidence specificity) and any external items, with URL and access date. Recommendations that rest on external research only carry basis = external.[1]
- External evidence vintage: accessed 2026-11-06; 256 items (partial 197, unverified 18, verified 41).[1]

**Where the findings are weak or counterintuitive**

- Joint bets resting on a node need inherited from its cell (a proxy): Natural graphite; Rare-earth oxides & compounds; Copper smelting & refining; Polysilicon.
- Evidence-led compacts (OSI cannot see the partner's capability in trade data): Grid Equipment & Conductors; Nuclear Fuel & Reactor Systems.
- Korea has no listed trade position (under 1% of world supply) at nodes where the US has material demand: rare-earth oxides & compounds, rare-earth permanent magnets, pv modules & generators, uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), uranium

conversion & enrichment, nuclear reactors & enrichment machinery, cobalt intermediates & chemicals, lithium chemicals, manganese dioxide (emd), piezo-electric quartz, solar thermal water heaters, insulators & bushings, electricity meters, hydro turbine parts, graphite electrodes, blade core wood (balsa), nuclear special materials, aluminium wire rod, direct-reduced iron (dri/hbi), pgms (iridium, ruthenium). Trade data miss capability that is owned abroad or sold at home, so these nodes cannot show complementarity even where corporate evidence suggests it.

- Low-evidence units among the active states: none.
- Fuel-trade lanes reach tier 1 (oil extraction; buyer: Korea): real energy-security lanes, but outside the electro-industrial compacts and with no market failure beyond the supply concentration the engine measures.

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Sources


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## Appendix — evidence tables and method

The tables below carry the full complementarity map, the node profiles and the trade actions that the main text summarises. Method, decision rules and limitations: docs/partnership_memos.md. Machine-readable outputs: country_profiles.csv, dyad_complementarity.csv, compact_candidates.csv, trade_priorities.csv, investment_priorities.csv, third_country_opportunities.csv, deal_pipeline.csv and evidence_register.csv.

### Table A1. Section 2, What Each Country Is Good At — and What It Needs

| COMPACT | NODE | STAGE | US SHARE | US NEED | KOREA SHARE | KOREA NEED | AIB CLASS | COERCIVE TOP SUPPLIER |
|---|---|---|---|---|---|---|---|---|
| SC-BATTERIES | Natural graphite | Extraction | 3.1% | high 0.82 | 2.6% | high 1.00 | BUILD | CHN 82% |
| SC-BATTERIES | Battery recycling feed | Recovery | 43.7% | low (cell 0.88; not a material buyer) | 0.3% | high 0.94 | LOW INTERVENTION | CHN 8% |
| SC-BATTERIES | Lithium minerals | Extraction | 4.0% | low (cell 0.82; not a material buyer) | 0.3% | high 1.00 | LOW INTERVENTION | CHN 25% |
| SC-BATTERIES | Cobalt intermediates & chemicals | Refining | 2.4% | demand 0.45 | 0.8% | high 0.99 | BUILD | CHN 76% |
| SC-BATTERIES | Lithium chemicals | Refining | 3.1% | demand 0.45 | 0.7% | high 0.99 | BUILD | CHN 74% |
| SC-BATTERIES | Manganese dioxide (EMD) | Refining | 3.7% | demand 0.45 | 0.0% | low (cell 0.99; not a material buyer) | BUILD | CHN 37% |
| SC-BATTERIES | Accumulator parts & separators | Components | 3.1% | high 0.73 | 12.7% | low (cell 0.97; not a material buyer) | BUILD | CHN 38% |
| SC-BATTERIES | Cathode materials & precursors | Refining | 4.9% | low 0.45 | 50.9% | low (cell 0.99; not a material buyer) | BUILD | CHN 92% |
| SC-BATTERIES | Synthetic graphite (anode-grade) | Refining | 5.7% | demand 0.45 | 4.6% | high 0.99 | BUILD | CHN 94% |
| SC-BATTERIES | Battery materials (foil, electrolyte) | Refining | 2.8% | demand 0.45 | 9.5% | high 0.99 | BUILD | CHN 16% |
| SC-BATTERIES | Li-ion cells & packs | Assembly | 6.4% | high 0.90 | 4.0% | low (cell 0.97; not a material buyer) | BUILD | CHN 84% |
| SC-BATTERIES | Other accumulators | Assembly | 7.4% | high 0.90 | 1.1% | low (cell 0.83; not a material buyer) | BUILD | CHN 28% |
| SC-EV | Electric buses, trucks & tractors | Assembly | 14.6% | low 0.15 | 0.4% | demand 0.37 | BUILD | CHN 24% |
| SC-EV | Electric rail vehicles | Assembly | 0.5% | low n/a | 10.6% | low n/a | LOW INTERVENTION | CHN 14% |
| SC-EV | Electric & plug-in cars | Assembly | 7.1% | demand 0.15 | 7.0% | low 0.37 | BUILD | CHN 69% |
| SC-EV | Electric two-wheelers & works trucks | Assembly | 5.4% | demand 0.15 | 1.1% | low 0.37 | BUILD | CHN 30% |
| SC-GRID | Copper ores & concentrates | Extraction | 4.7% | low (cell 0.89; not a material buyer) | 0.0% | high 0.97 | LOW INTERVENTION | CHN 13% |
| SC-GRID | Copper smelting & refining | Refining | 1.4% | high 0.90 | 1.7% | high 0.96 | LOW INTERVENTION | CHN 50% |
| SC-GRID | Grain-oriented electrical steel (GOES) | Refining | 3.7% | low (cell 0.90; not a material buyer) | 5.2% | low (cell 0.96; not a material buyer) | BUILD | CHN 41% |
| SC-GRID | Copper wire & rod | Refining | 4.9% | high 0.96 | 1.3% | low (cell 0.96; not a material buyer) | LOW INTERVENTION | CHN 6% |

| COMPACT | NODE | STAGE | US SHARE | US NEED | KOREA SHARE | KOREA NEED | AIB CLASS | COERCIVE TOP SUPPLIER |
|---|---|---|---|---|---|---|---|---|
| SC-GRID | Aluminium wire rod | Refining | 0.1% | demand n/a | 0.0% | not measured n/a | LOW INTERVENTION | RUS 7% |
| SC-GRID | Insulators & bushings | Components | 6.6% | demand 0.32 | 0.8% | low (cell 0.80; not a material buyer) | BUILD | CHN 32% |
| SC-GRID | Electricity meters | Assembly | 6.3% | demand 0.33 | 0.0% | low 0.64 | BUILD | CHN 34% |
| SC-GRID | Transformers | Assembly | 2.0% | demand 0.33 | 9.8% | low 0.64 | LOW INTERVENTION | CHN 21% |
| SC-GRID | Switchgear & protection | Assembly | 5.4% | demand 0.33 | 3.7% | low 0.64 | LOW INTERVENTION | CHN 20% |
| SC-GRID | Overhead conductors | Components | 1.0% | demand 0.33 | 1.2% | low 0.64 | LOW INTERVENTION | CHN 24% |
| SC-GRID | Power cables | Components | 6.7% | demand 0.33 | 8.4% | low 0.64 | LOW INTERVENTION | CHN 19% |
| SC-HEAT | Refrigerants | Refining | 19.9% | low n/a | 0.1% | demand n/a | BUILD | CHN 45% |
| SC-HEAT | Refrigeration & heat-pump compressors | Components | 7.6% | demand n/a | 7.0% | low n/a | BUILD | CHN 33% |
| SC-HEAT | Heat pumps & AC units | Assembly | 4.0% | demand n/a | 1.9% | low n/a | BUILD | CHN 49% |
| SC-HEAT | Solar thermal water heaters | Assembly | 0.4% | demand n/a | 0.0% | not measured n/a | BUILD | CHN 45% |
| SC-INDUSTRY | PGMs (iridium, ruthenium) | Refining | 4.0% | demand n/a | 0.0% | not measured n/a | LOW INTERVENTION | RUS 30% |
| SC-INDUSTRY | Electrolysers | Assembly | 7.5% | demand n/a | 5.7% | low n/a | BUILD | CHN 32% |
| SC-INDUSTRY | Hydrogen, ammonia & methanol | Refining | 8.7% | low n/a | 0.2% | demand n/a | LOW INTERVENTION | RUS 3% |
| SC-INDUSTRY | Furnace parts | Components | 7.7% | demand n/a | 2.2% | low n/a | BUILD | CHN 21% |
| SC-INDUSTRY | Graphite electrodes | Components | 2.3% | demand n/a | 0.0% | demand n/a | COORDINATE | CHN 38% |
| SC-INDUSTRY | Direct-reduced iron (DRI/HBI) | Refining | 5.2% | demand n/a | 0.0% | demand n/a | LOW INTERVENTION | RUS 20% |
| SC-INDUSTRY | Electric furnaces & heating | Assembly | 8.0% | demand n/a | 5.3% | low n/a | BUILD | CHN 27% |
| SC-MAGNETS | Rare-earth ores (monazite) | Extraction | 9.2% | low (cell 0.82; not a material buyer) | 0.0% | low (cell 1.00; not a material buyer) | INSUFFICIENT DATA | CHN 59% |
| SC-MAGNETS | Rare-earth oxides & compounds | Refining | 5.3% | high 0.96 | 1.0% | high 0.99 | BUILD | CHN 84% |
| SC-MAGNETS | Rare-earth metals | Refining | 5.3% | low (cell 0.96; not a material buyer) | 0.2% | low (cell 0.99; not a material buyer) | BUILD | CHN 84% |
| SC-MAGNETS | Rare-earth permanent magnets | Components | 1.8% | high 0.73 | 0.9% | high 0.97 | BUILD | CHN 64% |
| SC-MAGNETS | Non-oriented electrical steel (NGO) | Refining | 1.4% | low (cell 0.90; not a material buyer) | 13.7% | low (cell 0.96; not a material buyer) | BUILD | CHN 26% |
| SC-MAGNETS | Ferrite & other magnets | Components | 2.8% | high 0.73 | 4.4% | low (cell 0.97; not a material buyer) | BUILD | CHN 58% |
| SC-MAGNETS | Motor & generator parts | Components | 6.5% | high 0.73 | 2.0% | low (cell 0.97; not a material buyer) | BUILD | CHN 26% |

| COMPACT | NODE | STAGE | US SHARE | US NEED | KOREA SHARE | KOREA NEED | AIB CLASS | COERCIVE TOP SUPPLIER |
|---|---|---|---|---|---|---|---|---|
| SC-MAGNETS | AC generators | Assembly | 27.4% | low 0.27 | 0.3% | high 0.80 | LOW INTERVENTION | CHN 15% |
| SC-MAGNETS | Electric motors | Assembly | 5.4% | high 0.73 | 2.6% | high 0.97 | BUILD | CHN 26% |
| SC-NUCLEAR | Uranium/fuel Extraction: aggregated low-coverage sub-sectors (Spent fuel & nuclear scrap, Uranium ores) | Extraction | 0.7% | high 0.99 | 0.0% | low (cell 0.96; not a material buyer) | INSUFFICIENT DATA | RUS 8% |
| SC-NUCLEAR | Uranium conversion & enrichment | Refining | 3.3% | high 0.99 | 0.0% | high 0.98 | LOW INTERVENTION | RUS 16% |
| SC-NUCLEAR | Fuel assemblies & zirconium | Components | 9.2% | low 0.48 | 7.4% | low 0.54 | LOW INTERVENTION | RUS 28% |
| SC-NUCLEAR | Nuclear special materials | Refining | 3.5% | demand n/a | 0.1% | demand n/a | LOW INTERVENTION | 0% |
| SC-NUCLEAR | Reactor parts | Components | 18.1% | low 0.48 | 0.0% | low 0.54 | LOW INTERVENTION | RUS 45% |
| SC-NUCLEAR | Nuclear reactors & enrichment machinery | Assembly | 0.7% | demand 0.47 | 0.0% | low 0.62 | LOW INTERVENTION | RUS 49% |
| SC-SEMIS | Gallium, germanium & indium | Refining | 6.5% | demand n/a | 18.0% | low n/a | BUILD | CHN 25% |
| SC-SEMIS | Piezo-electric quartz | Refining | 3.3% | demand n/a | 0.0% | not measured n/a | BUILD | CHN 60% |
| SC-SEMIS | Semiconductor manufacturing equipment | Assembly | 15.6% | low 0.20 | 5.6% | demand 0.25 | LOW INTERVENTION | CHN 5% |
| SC-SEMIS | Sensors & controller parts | Components | 10.1% | low n/a | 2.4% | low n/a | BUILD | CHN 24% |
| SC-SEMIS | Integrated circuits | Components | 3.8% | low n/a | 17.1% | low n/a | LOW INTERVENTION | CHN 16% |
| SC-SEMIS | Power semiconductors | Components | 2.5% | demand 0.32 | 3.2% | low (cell 0.70; not a material buyer) | BUILD | CHN 32% |
| SC-SEMIS | Power electronics components | Components | 2.8% | high 0.73 | 2.5% | low (cell 0.97; not a material buyer) | BUILD | CHN 37% |
| SC-SEMIS | Automatic controllers | Assembly | 15.6% | low 0.28 | 2.8% | demand 0.51 | LOW INTERVENTION | CHN 12% |
| SC-SEMIS | Static converters & inverters | Assembly | 4.3% | demand 0.28 | 1.3% | low 0.51 | BUILD | CHN 38% |
| SC-SOLAR | Polysilicon | Refining | 30.7% | low (cell 0.71; not a material buyer) | 1.2% | high 0.71 | LOW INTERVENTION | CHN 97% |
| SC-SOLAR | Silver powder | Refining | 25.8% | low n/a | 1.6% | demand n/a | LOW INTERVENTION | 0% |
| SC-SOLAR | Wafers | Refining | 9.9% | low 0.21 | 6.4% | demand 0.30 | BUILD | CHN 98% |
| SC-SOLAR | Solar cells | Components | 0.9% | demand 0.21 | 11.1% | low 0.30 | BUILD | CHN 92% |
| SC-SOLAR | PV cover glass | Components | 2.5% | demand 0.21 | 1.9% | demand 0.30 | BUILD | CHN 61% |
| SC-SOLAR | PV modules & generators | Assembly | 4.5% | demand 0.20 | 0.0% | low 0.25 | BUILD | CHN 84% |
| SC-WIND | Blade core wood (balsa) | Refining | 0.1% | demand n/a | 0.0% | not measured n/a | LOW INTERVENTION | CHN 3% |
| SC-WIND | Wind blades & hubs | Components | 8.2% | demand 0.27 | 1.1% | low (cell 0.80; not a material buyer) | BUILD | CHN 71% |
| SC-WIND | Wind towers | Components | 0.4% | demand 0.27 | 11.7% | low (cell 0.80; not a material buyer) | LOW INTERVENTION | CHN 18% |
| SC-WIND | Hydro turbine parts | Components | 2.0% | demand n/a | 0.0% | not measured n/a | COORDINATE | CHN 35% |

| COMPACT | NODE | STAGE | US SHARE | US NEED | KOREA SHARE | KOREA NEED | AIB CLASS | COERCIVE TOP SUPPLIER |
|---|---|---|---|---|---|---|---|---|
| SC-WIND | Wind turbine generating sets | Assembly | 7.3% | low 0.33 | 0.1% | low 0.64 | BUILD | CHN 74% |
| SC-WIND | Hydro turbines | Assembly | 3.1% | low n/a | 0.0% | not measured n/a | LOW INTERVENTION | CHN 21% |

Sources: 1.

## Table A2. Section 3, Bilateral Complementarity Map

| RANK | TIER | TECHNOLOGY | UNIT | STAGE | US POSITION | KOREA POSITION | DYADIC RELATIONSHIP | STRATEGIC CASE |
|---|---|---|---|---|---|---|---|---|
| 1 | joint bet | Solar Manufacturing | Polysilicon | Refining | strong (30.7% of world supply) | moderate (1.2% of world supply) | US supplies / Korea needs | geopolitical chokepoint; need 0.71 / 0.71 |
| 2 | joint bet | Electric Vehicles | Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds) | Refining | moderate | weak | Capacity-constrained trusted supply | geopolitical chokepoint; need 0.90 / 0.99 |
| 3 | joint bet | Wind | Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds) | Refining | moderate | weak | Shared vulnerability | geopolitical chokepoint; need 0.96 / 0.75 |
| 4 | joint bet | Permanent Magnets & Drive Systems | Rare-earth oxides & compounds | Refining | moderate (5.3% of world supply) | weak (1.0% of world supply) | Capacity-constrained trusted supply | geopolitical chokepoint; need 0.96 / 0.99 |
| 5 | joint bet | Batteries | Batteries extraction (cobalt ores, lithium minerals, natural graphite) | Extraction | weak | weak | Shared vulnerability | geopolitical chokepoint; need 0.81 / 1.00 |
| 6 | joint bet | Battery Materials & Cells | Natural graphite | Extraction | moderate (3.1% of world supply) | moderate (2.6% of world supply) | Shared vulnerability | geopolitical chokepoint; need 0.82 / 1.00 |
| 7 | joint bet | Nuclear | Nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores)) | Extraction | weak | weak | Shared vulnerability | geopolitical chokepoint; need 0.99 / 0.96 |
| 8 | joint bet | Battery Materials & Cells | Battery recycling feed | Recovery | strong (43.7% of world supply) | weak (0.3% of world supply) | US supplies / Korea needs | emerging bottleneck; need 0.88 / 0.94 |
| 9 | joint bet | Permanent Magnets & Drive Systems | Rare-earth permanent magnets | Components | moderate (1.8% of world supply) | weak (0.9% of world supply) | Shared vulnerability | geopolitical chokepoint; need 0.73 / 0.97 |
| 10 | joint bet | Solar | Solar refining (copper wire & rod, polysilicon) | Refining | weak | weak | Shared vulnerability | geopolitical chokepoint; need 0.71 / 0.71 |
| 11 | joint bet | Wind | Wind components (ac generators, ferrite & other magnets, insulators & bushings) | Components | strong | moderate | US supplies / Korea needs | strategic manufacturing gap; need 0.27 / 0.80 |
| 12 | joint bet | Battery Materials & Cells | Accumulator parts & separators | Components | moderate (3.1% of world supply) | strong (12.7% of world supply) | Capacity-constrained trusted supply | trusted partner capacity constraint; need 0.73 / 0.97 |
| 13 | joint bet | Batteries | Batteries components (accumulator parts & separators) | Components | strong | moderate | US supplies / Korea needs | strategic manufacturing gap; need 0.06 / 0.73 |
| 14 | joint bet | Electric Grid | Electric Grid refining (copper smelting & refining, copper wire & rod, grain-oriented electrical steel (goes)) | Refining | weak | weak | Shared vulnerability | geopolitical chokepoint; need 0.90 / 0.96 |
| 15 | joint bet | Grid Equipment & Conductors | Copper smelting & refining | Refining | weak (1.4% of world supply) | weak (1.7% of world supply) | Shared vulnerability | geopolitical chokepoint; need 0.90 / 0.96 |
| 16 | joint bet | Permanent Magnets & Drive Systems | AC generators | Assembly | strong (27.4% of world supply) | weak (0.3% of world supply) | US supplies / Korea needs | strategic manufacturing gap; need 0.27 / 0.80 |
| 17 | joint bet | Oil | Oil extraction | Extraction | strong | weak | Capacity-constrained trusted supply | trusted partner capacity constraint; need 0.47 / 0.99 |
| 18 | coordinate | Solar Manufacturing | Solar cells | Components | moderate (0.9% of world supply) | strong (11.1% of world supply) | Capacity-constrained trusted supply | geopolitical chokepoint; need 0.21 / 0.30 |
| 19 | coordinate | Battery Materials & Cells | Cathode materials & precursors | Refining | moderate (4.9% of world supply) | strong (50.9% of world supply) | Complementary specialisation | geopolitical chokepoint; need 0.45 / 0.99 |

| RANK | TIER | TECHNOLOGY | UNIT | STAGE | US POSITION | KOREA POSITION | DYADIC RELATIONSHIP | STRATEGIC CASE |
|---|---|---|---|---|---|---|---|---|
| 20 | coordinate | Electric Vehicles | Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite) | Extraction | moderate | weak | Capacity-constrained trusted supply | geopolitical chokepoint; need 0.82 / 1.00 |
| 21 | coordinate | Electric Vehicles | Electric buses, trucks & tractors | Assembly | strong (14.6% of world supply) | weak (0.4% of world supply) | Capacity-constrained trusted supply | trusted partner capacity constraint; need 0.15 / 0.37 |
| 22 | coordinate | Gas | Gas assembly | Assembly | moderate | strong | Complementary specialisation | strategic manufacturing gap; need 0.18 / 0.34 |
| 23 | coordinate | Gas | Gas components | Components | strong | moderate | Complementary specialisation | strategic manufacturing gap; need 0.03 / 0.50 |
| 24 | coordinate | Gas | Gas extraction | Extraction | strong | weak | US supplies / Korea needs | no material intervention; need 0.30 / 0.96 |
| 25 | coordinate | Gas | Gas refining | Refining | strong | weak | US supplies / Korea needs | no material intervention; need 0.09 / 0.93 |
| 26 | coordinate | Heat Pumps & Cooling | Refrigerants | Refining | strong (19.9% of world supply) | weak (0.1% of world supply) | Capacity-constrained trusted supply | trusted partner capacity constraint; need n/a / n/a |
| 27 | coordinate | Semiconductors & Power Electronics | Gallium, germanium & indium | Refining | moderate (6.5% of world supply) | strong (18.0% of world supply) | Capacity-constrained trusted supply | trusted partner capacity constraint; need n/a / n/a |
| 28 | coordinate | Semiconductors & Power Electronics | Sensors & controller parts | Components | strong (10.1% of world supply) | moderate (2.4% of world supply) | Complementary specialisation | trusted partner capacity constraint; need n/a / n/a |
| 29 | coordinate | Permanent Magnets & Drive Systems | Non-oriented electrical steel (NGO) | Refining | moderate (1.4% of world supply) | strong (13.7% of world supply) | Complementary specialisation | trusted partner capacity constraint; need 0.90 / 0.96 |
| 30 | coordinate | Semiconductors & Power Electronics | Automatic controllers | Assembly | strong (15.6% of world supply) | moderate (2.8% of world supply) | Complementary specialisation | strategic manufacturing gap; need 0.28 / 0.51 |
| 31 | coordinate | Semiconductors & Power Electronics | Integrated circuits | Components | moderate (3.8% of world supply) | strong (17.1% of world supply) | Complementary specialisation | strategic manufacturing gap; need n/a / n/a |
| 32 | coordinate | Semiconductors & Power Electronics | Semiconductor manufacturing equipment | Assembly | strong (15.6% of world supply) | moderate (5.6% of world supply) | Complementary specialisation | strategic manufacturing gap; need 0.20 / 0.25 |
| 33 | coordinate | Solar Manufacturing | Silver powder | Refining | strong (25.8% of world supply) | weak (1.6% of world supply) | Complementary specialisation | strategic manufacturing gap; need n/a / n/a |
| 34 | coordinate | Oil | Oil assembly (ac generators) | Assembly | moderate | strong | Complementary specialisation | strategic manufacturing gap; need 0.14 / 0.30 |
| 35 | coordinate | Electric Vehicles | Electric rail vehicles | Assembly | weak (0.5% of world supply) | strong (10.6% of world supply) | Complementary specialisation | strategic manufacturing gap; need n/a / n/a |
| 36 | coordinate | Coal | Coal components | Components | moderate | strong | Complementary specialisation | strategic manufacturing gap; need 0.49 / 0.58 |
| 37 | coordinate | Coal | Coal assembly | Assembly | moderate | strong | Complementary specialisation | strategic manufacturing gap; need 0.42 / 0.60 |
| 38 | coordinate | Oil | Oil refining | Refining | strong | moderate | Complementary specialisation | strategic manufacturing gap; need 0.00 / 0.23 |
| 39 | coordinate | Coal | Coal extraction | Extraction | strong | weak | US supplies / Korea needs | no material intervention; need 0.61 / 0.94 |
| 40 | manage competition | Electric Vehicles | Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) | Components | strong | strong | Direct competition | no material intervention; need 0.73 / 0.97 |
| 41 | manage competition | Electric Vehicles | Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks) | Assembly | strong | strong | Direct competition | no material intervention; need 0.15 / 0.37 |
| 42 | manage competition | Batteries | Batteries assembly (li-ion cells & packs, other accumulators) | Assembly | strong | strong | Direct competition | geopolitical chokepoint; need 0.90 / 0.83 |

Sources: 1.

## Table A3. Section 4, Trade Agenda

| ID | DIRECTION | UNIT | ACTION TYPE | ACTION |
|---|---|---|---|---|
| T01 | USA -> KOR | polysilicon | predictability | Keep US exports of Polysilicon to Korea tariff-free and predictable; the buyer's need is met by the trusted supplier. |
| T02 | USA -> KOR | Electric Vehicles/Refining | procurement recognition | Recognise US output of Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds) as trusted domestic-equivalent supply in Korea procurement and content rules; capacity, not access, is the constraint. |
| T03 | both | Wind/Refining | common standards | Common specifications for Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds) so a joint supplier base can serve both markets. |
| T04 | USA -> KOR | rare-earth-oxides-and-compounds | procurement recognition | Recognise US output of Rare-earth oxides & compounds as trusted domestic-equivalent supply in Korea procurement and content rules; capacity, not access, is the constraint. |
| T05 | both | Batteries/Extraction | common standards | Common specifications for Batteries extraction (cobalt ores, lithium minerals, natural graphite) so a joint supplier base can serve both markets. |
| T06 | both | natural-graphite | common standards | Common specifications for Natural graphite so a joint supplier base can serve both markets. |
| T07 | both | Nuclear/Extraction | common standards | Common specifications for Nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores)) so a joint supplier base can serve both markets. |
| T08 | USA -> KOR | battery-recycling-feed | predictability | Keep US exports of Battery recycling feed to Korea tariff-free and predictable; the buyer's need is met by the trusted supplier. |
| T09 | both | rare-earth-permanent-magnets | anti circumvention | Common origin and anti-circumvention rules for Rare-earth permanent magnets: both import heavily from CHN, so a measure one side takes alone leaks through the other. |
| T10 | both | Solar/Refining | common standards | Common specifications for Solar refining (copper wire & rod, polysilicon) so a joint supplier base can serve both markets. |
| T11 | USA -> KOR | Wind/Components | tariff risk | Bind market access for US Wind components (ac generators, ferrite & other magnets, insulators & bushings) into Korea: Korea's protection priority on this node is in its top third (0.88), which taxes the trusted supplier it needs. |
| T12 | KOR -> USA | accumulator-parts-and-separators | procurement recognition | Recognise Korea output of Accumulator parts & separators as trusted domestic-equivalent supply in US procurement and content rules; capacity, not access, is the constraint. |
| T13 | open | uranium-fuel-upstream--aggregated | open sourcing | Leave Uranium/fuel Extraction: aggregated low-coverage sub-sectors (Spent fuel & nuclear scrap, Uranium ores) open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security. |
| T14 | open | nuclear-special-materials | open sourcing | Leave Nuclear special materials open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security. |
| T15 | open | transformers | open sourcing | Leave Transformers open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security. |
| T16 | open | uranium-conversion-and-enrichment | open sourcing | Leave Uranium conversion & enrichment open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security. |
| T17 | open | overhead-conductors | open sourcing | Leave Overhead conductors open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security. |
| T18 | open | wind-towers | open sourcing | Leave Wind towers open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security. |
| T19 | both | Electric Vehicles/Components | no preference | No bilateral preference for Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets): both are strong suppliers; keep the market contestable. |
| T20 | both | Electric Vehicles/Assembly | no preference | No bilateral preference for Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks): both are strong suppliers; keep the market contestable. |
| T21 | both | Batteries/Assembly | no preference | No bilateral preference for Batteries assembly (li-ion cells & packs, other accumulators): both are strong suppliers; keep the market contestable. |

Sources: 1.

*Rule-generated from the data (R/partnerships) with external facts from a dated, curated evidence layer; not yet checked by an independent verifier. Items marked partial or unverified must be confirmed before citation.*